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AI25BTECH11034 - Sujal Chauhan

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Question

Q. The matrix

$$A = \begin{pmatrix} \frac{3}{2} & 0 & \frac{1}{2} \\ 0 & -1 & 0 \\ \frac{1}{2} & 0 & \frac{3}{2} \end{pmatrix}$$

has three distinct eigenvalues and one of its eigenvectors is $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$. Which one of the following can be another eigenvector of A ?

- (a) $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ (b) $\begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}$ (c) $\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$ (d) $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

Theory

Let A be a symmetric matrix ($A^T = A$). Suppose $\mathbf{a}$ and $\mathbf{b}$ are eigenvectors of A corresponding to distinct eigenvalues λ_1 and λ_2 respectively. That is:

$$A\mathbf{a} = \lambda_1\mathbf{a}, \quad A\mathbf{b} = \lambda_2\mathbf{b}$$

Consider the product $\mathbf{a}^T A \mathbf{b}$ two ways:

$$\mathbf{a}^T A \mathbf{b} = \mathbf{a}^T (\lambda_2 \mathbf{b}) = \lambda_2 \mathbf{a}^T \mathbf{b}$$

But since A is symmetric,

$$\mathbf{a}^T A \mathbf{b} = (A^T \mathbf{a})^T \mathbf{b} = (A \mathbf{a})^T \mathbf{b} = (\lambda_1 \mathbf{a})^T \mathbf{b} = \lambda_1 (\mathbf{a}^T \mathbf{b})$$

Equating the two results:

$$\lambda_1(\mathbf{a}^T \mathbf{b}) = \lambda_2(\mathbf{a}^T \mathbf{b})$$

Since $\lambda_1 \neq \lambda_2$, it follows that

$$(\lambda_1 - \lambda_2)(\mathbf{a}^T \mathbf{b}) = 0$$

$$\implies \mathbf{a}^T \mathbf{b} = 0$$

Thus, the eigenvectors $\mathbf{a}$ and $\mathbf{b}$ are perpendicular.

Solution

Let

$$A = \begin{pmatrix} \frac{3}{2} & 0 & \frac{1}{2} \\ 0 & -1 & 0 \\ \frac{1}{2} & 0 & \frac{3}{2} \end{pmatrix}, \quad a = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

Since A is symmetric, eigenvectors corresponding to distinct eigenvalues are orthogonal. Compute the inner product $a^T b$ for each choice b :

Solution

$$(a) \quad b = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} : \quad a^T b = 1 \cdot 1 + 0 \cdot 0 + 1 \cdot (-1) = 0,$$

$$(b) \quad b = \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix} : \quad a^T b = -1 \neq 0,$$

$$(c) \quad b = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} : \quad a^T b = -1 \neq 0,$$

$$(d) \quad b = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} : \quad a^T b = 1 \neq 0.$$

Solution

Only (a) is orthogonal to a ; we now verify that it is indeed an eigenvector by direct multiplication:

$$A \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} \frac{3}{2} \cdot 1 + \frac{1}{2} \cdot (-1) \\ 0 \\ \frac{1}{2} \cdot 1 + \frac{3}{2} \cdot (-1) \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}.$$

Hence $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ is an eigenvector (with eigenvalue 1), so the correct choice is (a).